Hose Pump



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: S-hose pump

What it is used for

To operate an S-hose pump for transferring compatible liquids using the pump's moving hose and compression mechanism. To maintain steady flow while preventing hose damage, leakage, and unsafe exposure to the pumped fluid.

Main parts

- Pump casing and drive motor
- Rotor or hose-compression assembly
- Pump hose or tube
- Inlet and outlet connections
- Speed controller and power switch
- Mounting frame and protective cover

Before you start

1. Read the pump manual and confirm that the hose material is compatible with the liquid.
2. Inspect the hose, fittings, casing, drive, and power cable for wear or damage.
3. Check that the pump is mounted securely and the protective cover is fitted.
4. Verify that the inlet and outlet lines are correctly connected and not kinked or blocked.
5. Confirm that the liquid temperature, viscosity, and chemical properties are within the pump's rated limits.
6. Ensure the discharge line is open and the receiving container has enough capacity.

How to use

1. Switch off and isolate the pump before connecting or adjusting hoses.
2. Secure the inlet and outlet hoses and check that all connections are leak-tight.
3. Place the inlet line in the source liquid and the outlet line in the receiving vessel as required.
4. Confirm that the pump is primed if the model or process requires priming.
5. Set the speed controller to a low starting setting.
6. Switch on the pump and check the direction of flow and initial delivery.
7. Increase the speed gradually to the required flow rate within the rated operating range.
8. Monitor the hose, connections, motor, and discharge for leakage, abnormal vibration, or overheating.
9. Adjust the flow only using the approved controller; do not squeeze or obstruct the hose by hand.
10. Stop the pump if the hose collapses, leaks, slips, or the flow becomes irregular.
11. Switch off and isolate the power before removing the hose or clearing a blockage.

2. Drain and flush the pump with a compatible cleaning fluid if required by the process.
3. Clean the external surfaces and inspect the hose before returning the pump to service.

⚠ Safety

- Operate only within the manufacturer’s pressure, speed, temperature, and chemical-compatibility limits.
- Never operate with a missing cover or reach into the moving rotor or hose-compression area.
- Do not run the pump dry unless the manufacturer explicitly permits it.
- Depressurize and isolate the system before disconnecting any hose or fitting.
- Use suitable gloves, eye protection, and other PPE for the pumped liquid.
- Keep electrical connections dry and use appropriate protection when pumping hazardous or conductive liquids.

Care & storage

- Inspect the hose regularly for flattening, cracks, swelling, hardening, or wear.
- Replace the hose at the recommended interval or when damage appears.
- Clean and flush fluid-contact parts using compatible materials and approved procedures.
- Check mounting bolts, fittings, and drive components for looseness or wear.
- Keep the pump dry, clean, and protected during storage.

Common mistakes to avoid

- Using a hose material incompatible with the pumped liquid.
- Running the pump dry or against a blocked discharge line.
- Operating above the rated speed or pressure.
- Opening the pump while the drive is energized or the line is pressurized.
- Ignoring hose wear, leaks, or abnormal vibration.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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